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METHOD OF PROGRAMMING A LOAD CONTROL DEVICE

Abstract

A load control device is able to receive radio-frequency (RF) signals from a Wi-Fi-enabled device, such as a smart phone, via a wireless local area network. The load control device comprises a controllably conductive device adapted to be coupled in series between an AC power source and an electrical load, a controller for rendering the controllably conductive device conductive and non-conductive, and a Wi-Fi module operable to receive the RF signals from the wireless network. The controller controls the controllably conductive device to adjust the power delivered to the load in response to the wireless signals received from the wireless network. The load control device may further comprise an optical module operable to receive an optical signal, such that the controller may obtain an IP address from the received optical signal and control the power delivered to the load in response to a wireless signal that includes the IP address.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/769,831, filed Jul. 11, 2024; which is a continuation of U.S. patent application Ser. No. 17/861,349, filed Jul. 11, 2022, now U.S. Pat. No. 12,075,321 issued Aug. 27, 2024; which is a continuation of U.S. patent application Ser. No. 17/013,108, filed Sep. 4, 2020, now U.S. Pat. No. 11,388,570 issued Jul. 12, 2022; which is a continuation of U.S. patent application Ser. No. 15/352,673 filed Nov. 16, 2016, now U.S. Pat. No. 10,779,381 issued Sep. 15, 2020; which is a continuation of U.S. patent application Ser. No. 13/538,615, filed Jun. 29, 2012, now U.S. Pat. No. 9,544,977 issued Jun. 30, 2011; which claims the benefit of U.S. Provisional Patent Application No. 61/503,292, filed on Jun. 30, 2011; all of which are hereby incorporated by reference herein in their entirety. [0002] This application is related to commonly assigned U.S. patent application Ser. No. 13/538,555, filed Jun. 29, 2012, entitled LOAD CONTROL DEVICE HAVING INTERNET CONNECTIVITY; and to commonly assigned U.S. patent application Ser. No. 13/538,665, filed Jun. 29, 2012, entitled METHOD OF OPTICALLY TRANSMITTING DIGITAL INFORMATION FROM A SMART PHONE TO A CONTROL DEVICE, the contents of each are hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0003] The present invention relates to a load control device for controlling the amount of power delivered to an electrical load, and more particularly, to a wall-mounted dimmer switch that is operable to connect to the Internet via a wireless connection and to control a lighting load in response to messages received via the Internet.

Description of the Related Art

[0004] A load control device may be adapted to be coupled in a series electrical connection between an alternating-current (AC) power source and an electrical load for control of the power delivered from the AC power source to the electrical load. Prior art load control devices include, for example, lighting control devices (such as wall-mounted dimmer switches and plug-in lamp dimmers), motor control devices (for motor loads), temperature control devices, motorized window treatments, and remote controls. Some load control devices are operable to transmit and receive wireless signals, such as radio-frequency (RF) or infrared (IR) signals, to thus provide for wireless control of the corresponding loads. One example of an RF lighting control system is disclosed in commonly-assigned U.S. Pat. No. 5,905,442, issued May 18, 1999, entitled METHOD AND APPARATUS FOR CONTROLLING AND DETERMINING THE STATUS OF ELECTRICAL DEVICES FROM REMOTE LOCATIONS, the entire disclosure of which is hereby incorporated by reference.

[0005] There is a need for a wireless load control device that is operable to connect to the Internet via a wireless connection and to control or program a lighting load in response to messages received from a wireless device (e.g., received via the Internet). It would be particularly desirable if such load control device were operable to be controlled or programmed from a Wi-Fi enabled control device, such as a smart phone (for example, an iPhone® or Android® smart phone).

SUMMARY OF THE INVENTION

[0006] According to an embodiment of the present invention, a load control device may be implemented to control power delivered from an AC power source to an electrical load. The load control device may include a controllably conductive device, a controller, and a near-field communications (NFC) module. The controllably conductive device may be adapted to be coupled in series electrical connection between the source and the load. The controller may be operatively coupled to a control input of the controllably conductive device for rendering the controllably conductive device conductive and non-conductive. The NFC module may be operatively coupled to the controller. The NFC may be operable to receive at least one NFC signal for programming operating parameters of the load control device. The controller may be operable to control the controllably conductive device to control the power delivered to the load based on the operating parameters of the load control device.

[0007] According to an embodiment of the present invention, the load control device may include a controllably conductive device, a controller, and an Internet Protocol communications module. The controllably conductive device may be adapted to be coupled in series electrical connection between the source and the load. The controller may be operatively coupled to the controllably conductive device for controlling the controllably conductive device. The Internet Protocol communications module may be operatively coupled to the controller. The Internet Protocol communications module may be operable to receive an Internet Protocol packet for programming the load control device. The controller may be operable to control the controllably conductive device based on the programming of the load control device.

[0008] According to an embodiment of the present invention, the load control device may include a controllably conductive device, a controller, and an optical module. The controllably conductive device may be adapted to be coupled in series electrical connection between the source and the load. The controller may be operatively coupled to a control input of the controllably conductive device for rendering the controllably conductive device conductive and non-conductive. The optical module may be operatively coupled to the controller. The optical module may be operable to receive an optical signal for programming operating parameters of the load control device. The controller may be operable to control the controllably conductive device to control the power delivered to the load based on the optical signal programming of the load control device.

[0009] According to an embodiment of the present invention, a load control system is described for controlling power delivered from an AC power source to an electrical load. The load control system may include a load control device and a smart phone. The load control device may be adapted to be coupled in series electrical connection between the source and the load for controlling the power delivered to the load. The load control device may store one or more operating parameters. The smart phone may include a visual display for providing a user interface for adjusting the one or more operating parameters of the load control device. The smart phone may be operable to directly transmit operating parameters to the load control device.

[0010] According to an embodiment of the present invention, a lighting control device may be programmed using a wireless control device. The wireless control device may include a visual display and/or a camera. The lighting control device may be operable to adjust the intensity of a lighting load. The lighting load may be purchased in packaging having a barcode. The barcode of the packaging of the lighting load may be scanned using the camera of the wireless control device. The wireless control device may determine an operating parameter for the lighting load using information received from the scanned barcode. A digital message may be transmitted that includes

the operating parameter to the lighting control device. The operating parameter may be stored in the lighting control device in response to the lighting control device receiving the digital message. [0011] According to an embodiment of the present invention, the lighting control device may be operable to adjust the intensity of a lighting load by transmitting a first digital message from a wireless control device when the camera of the wireless control device is directed at the lighting load. The intensity of the lighting load may be adjusted to a first intensity in response to the lighting control device receiving the first digital message. The intensity of the lighting load may be decreased from the first intensity while the camera of the wireless control device is directed at the lighting load. A second digital message may be transmitted from the wireless control device to the lighting control device when the wireless control device detects flickering of the intensity of the lighting load. The intensity of the lighting load may be increased in response to the lighting control device receiving the second digital message. A third digital message may be transmitted from the wireless control device to the lighting control device when the wireless control device does not detecting flickering in the intensity of the lighting control device. The adjustment of the intensity of the lighting load may be ceased in response to the lighting control device receiving the third digital message, such that the intensity of the lighting load is at a second intensity. The second intensity may be stored as a low-end intensity of the lighting control device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The invention will now be described in greater detail in the following detailed description with reference to the drawings in which:

[0013] FIG. 1 is a simple diagram of a radio-frequency (RF) lighting control system comprising a dimmer switch and a wireless control device, such as a smart phone, according to a first embodiment of the present invention;

[0014] FIG. 2 is an example screenshot that may be provided on the wireless control device for controlling the dimmer switch of the RF lighting control system of FIG. 1;

[0015] FIG. 3 is an example screenshot that may be provided on the wireless control device for programming the dimmer switch of the RF lighting control system of FIG. 1;

[0016] FIG. 4 is a simplified block diagram of the dimmer switch of the RF lighting control system of FIG. 1;

[0017] FIG. 5 is a front view of the wireless control device showing an example location of a portion of a display screen that may be used to transmit optical signals;

[0018] FIG. 6 is a perspective view showing the display screen of the wireless control device directed towards an optical receiver of the dimmer switch while the wireless control device is transmitting the optical signals;

[0019] FIG. 7 is a perspective view showing a camera lens and a camera flash light source of the wireless control device directed towards the optical receiver of the dimmer switch according to an example embodiment of the present invention;

[0020] FIG. 8 is a front view of the wireless control device showing an example targeting image for assisting in lining up the camera lens and the camera flash light source with the optical receiver of the dimmer switch according to an example embodiment of the present invention;

[0021] FIG. 9 is a simplified flow diagram illustrating an example embodiment for optically programming or controlling the dimmer switch via the wireless control device;

[0022] FIG. 10 is a simplified flow diagram illustrating an example embodiment for programming or controlling the dimmer switch via a Wi-Fi signal;

[0023] FIG. 11 is a simplified block diagram of a dimmer switch according to an alternate embodiment of the present invention;

[0024] FIG. **12** is a simplified flow diagram illustrating an example embodiment for programming or controlling the dimmer switch via near field communication (NFC) signals;
[0025] FIG. **13** is a simplified flow diagram illustrating another example embodiment for programming the dimmer switch via the NFC signals;
[0026] FIG. **14** is a simplified flow diagram illustrating another example embodiment for programming the dimmer switch;
[0027] FIG. **15** is a simple diagram of an RF lighting control system that using a proprietary protocol according to an alternate embodiment;
[0028] FIG. **16** is a simplified flow diagram illustrating an example embodiment for programming or controlling the dimmer switch using proprietary-protocol communications; and
[0029] FIG. **17** is a simplified flow diagram illustrating an example embodiment for programming the dimmer switch to corresponding low-end and high-end limits for a particular lamp.

DETAILED DESCRIPTION OF THE INVENTION

[0030] The foregoing summary, as well as the following detailed description of the preferred embodiments, is better understood when read in conjunction with the appended drawings. For the purposes of illustrating the invention, there is shown in the drawings an embodiment that is presently preferred, in which like numerals represent similar parts throughout the several views of the drawings, it being understood, however, that the invention is not limited to the specific methods and instrumentalities disclosed.

[0031] FIG. **1** is a simple diagram of a radio-frequency (RF) lighting control system **100** that includes a dimmer switch **110** and a wireless control device **120**, according to an example embodiment of the present invention. The wireless control device **120** may be any device capable of performing wireless communications, such as, a smart phone (for example, an iPhone® smart phone, an Android® smart phone, or a Blackberry® smart phone), a personal computer, a laptop, a wireless-capable media device (e.g., MP3 player, gaming device, or television), or a tablet device, (for example, an iPad® hand-held computing device), a Wi-Fi or wireless-communication-capable television, or any other suitable Internet-Protocol-enabled device.

[0032] According to an embodiment of the present invention, the wireless control device **120** may be operable to transmit digital messages in one or more Internet Protocol packets to the dimmer switch **110**. The Internet Protocol (IP) is responsible for addressing hosts and routing datagrams (i.e., packets) from a source host to a destination host across one or more IP networks. For this purpose, the Internet Protocol defines an addressing system that has two functions: identifying hosts and providing a logical location service. This is accomplished by defining standard datagrams and a standard addressing system.

[0033] Each datagram has two components, a header and a payload. The IP header is tagged with the source IP address, destination IP address, and other meta-data needed to route and deliver the datagram. The payload is the data to be transported.

[0034] The wireless control device **120** may transmit the digital messages via RF signals **106** either directly or via a wireless network that includes a standard wireless router **130**. For example, the wireless control device **120** may transmit the RF signals **106** directly to the dimmer switch **110** via a point-to-point communication, such as a Wi-Fi communication link, e.g., an 802.11 wireless local area network (LAN), or other direct wireless communication link, e.g., a Wi-MAX communication link or a Bluetooth® communication link. This point-to-point communication may be performed using a standardized communication, e.g., Wi-Fi Direct, or any non-standardized communication that allows a wireless device to connect to another wireless device without the use of a wireless access point. For example, the wireless control device **120** and/or the dimmer switch **110** may download a software access point (AP) that provides a protected wireless communication between the devices.

[0035] The wireless control device **120** may also transmit RF signals **106** to the dimmer switch **110** via a wireless network. The wireless network may enable wireless communications via one or more

wireless communications links, such as a Wi-Fi communications link, a Wi-MAX communications link, a Bluetooth® communications link, a cellular communications link, a television white space (TVWS) communication link, or any combination thereof. For example, the wireless control device **120** may communicate with a network server via a first wireless communications link (e.g., a cellular communications link), while the dimmer switch **110** communicates with the network server via a second communications link (e.g., a Wi-Fi communications link). In an alternative embodiment, the wireless control device **120** and the dimmer switch **110** may communicate with the network via the same type of communication link. The lighting control system **100** may also include a femtocell, a Home Node B, and/or other network entity for facilitating the configuration and operation of the lighting control system and for allowing wireless communications and connection to the Internet.

[0036] The dimmer switch **110** may be coupled in series electrical connection between an AC power source **102** and a lighting load **104** for controlling the amount of power delivered to the lighting load. The dimmer switch **110** may be wall-mounted in a standard electrical wallbox, or alternatively implemented as a table-top load control device. The dimmer switch **110** comprises a faceplate **112** and a bezel **113** received in an opening of the faceplate. The dimmer switch **110** further comprises a toggle actuator **114** and an intensity adjustment actuator **116**. Actuations of the toggle actuator **114** toggle, e.g., alternately turn off and on, the lighting load **104**. Actuations of an upper portion **116A** or a lower portion **116B** of the intensity adjustment actuator **116** may respectively increase or decrease the amount of power delivered to the lighting load **104** and thus increase or decrease the intensity of the lighting load **104** from a minimum (i.e., low-end) intensity (e.g., approximately 1-10%) to a maximum (i.e., high-end) intensity (e.g., approximately 100%). A plurality of visual indicators **118**, e.g., light-emitting diodes (LEDs), may be arranged in a linear array on the left side of the bezel **113**. The visual indicators **118** are illuminated to provide visual feedback of the intensity of the lighting load **104**. An example of a dimmer switch having a toggle actuator and an intensity adjustment actuator is described in greater detail in U.S. Pat. No. 5,248,919 (“the 919 patent”), issued Sep. 28, 1993, entitled LIGHTING CONTROL DEVICE, the entire disclosure of which is hereby incorporated by reference. Alternatively, the dimmer switch **110** could be replaced by an electronic switch for simply turning the lighting load **104** on and off. The electronic switch may include a single visual indicator, e.g., the middle indicator of the visual indicators **118** of the dimmer switch **110**.

[0037] According to an example embodiment of the present invention, the dimmer switch **110** may include an optical receiver **119**. The optical receiver **119** may be used to receive optical signals from the wireless control device **120**. Optical signals may be free-space optical communications or communications via physical connections. For example, free space optical communications may include communications via air, while physical optical communications may include communications via optical fiber cable or an optical transmission pipe. The optical signals may also be included in visible light, e.g., a flashing light, or non-visible light, e.g., infrared, spectrums.

[0038] The optical signals may provide instructions for programming and/or adjusting the operating parameters (e.g., the low-end intensity and the high-end intensity) of the dimmer switch **110**. For example, the optical signals may be used to configure the dimmer switch such that the dimmer switch **110** is operable to receive the RF signals **106** from the wireless control device **120** as will be described in greater detail below. The optical signals may also be used to control or program the lighting configurations of the dimmer switch **110**. And, though embodiments described herein may be described with respect to using optical signals or other signals to program or control a dimmer switch from a wireless control device, such signals may be used to program or control any device that is capable of receiving instructions via such optical or other signals, such as shades, thermostats, plug-in devices, or the like.

[0039] Wireless load control devices are described in greater detail in commonly-assigned U.S. Pat. No. 5,838,226, issued Nov. 17, 1998, entitled COMMUNICATION PROTOCOL FOR

TRANSMISSION SYSTEM FOR CONTROLLING AND DETERMINING THE STATUS OF ELECTRICAL DEVICES FROM REMOTE LOCATIONS; U.S. Pat. No. 6,803,728, issued Oct. 12, 2004, entitled SYSTEM FOR CONTROL OF DEVICES; U.S. patent application Ser. No. 12/033,223, filed Feb. 19, 2008, entitled COMMUNICATION PROTOCOL FOR A RADIO-FREQUENCY LOAD CONTROL SYSTEM; and U.S. patent application Ser. No. 13/234,573, filed Sep. 16, 2011, entitled DYNAMIC KEYPAD FOR CONTROLLING ENERGY-SAVINGS SETTINGS OF A LOAD CONTROL SYSTEM; the entire disclosures of which are hereby incorporated by reference.

[0040] The wireless control device **120** has a visual display **122**, which may comprise a touch screen having, for example, a capacitive touch pad displaced overtop the visual display, such that the visual display may display soft buttons that may be actuated by a user. Alternatively, the wireless control device **120** may comprise a plurality of hard buttons (e.g., physical buttons) in addition to the visual display **122**. The wireless control device **120** may download a product control application for allowing the user to control the lighting load **104**. In response to actuations of the displayed soft buttons or hard buttons, the wireless control device **120** transmits digital messages to the dimmer switch **110** directly or through other wireless communications described herein. For example, the digital messages may be transmitted via Wi-Fi communication using the wireless router **130**. The dimmer switch **110** may adjust the intensity of the lighting load **104** in response to commands included in the digital messages, such that the dimmer switch controls the lighting load in response to actuations of the soft buttons or hard buttons of the wireless control device **120**.

[0041] In addition, the wireless control device **120** may be controlled to transmit optical signals, near field communication (NFC) signals, or RF signals according to a proprietary RF communication protocol (such as, for example, the Clear Connect™ protocol) as described herein. For example, the visual display **122** may be controlled to transmit optical signals to the optical receiver **119** of the dimmer switch **110** (as will be described in greater detail below). The wireless control device **120** may also comprise a camera lens **124** (FIG. 6) and a camera flash lighting source **126** (FIG. 6), which may also be used to transmit and receive optical signals for controlling the lighting load.

[0042] The dimmer switch **110** and the wireless control device **120** may both be assigned a unique address for wireless communications described herein. For example, where wireless communications are performed using a Wi-Fi communication link, a Media Access Control (MAC) address may be assigned (e.g., during manufacture). The wireless control device **120** may connect to the wireless LAN via the wireless router **130** using standard procedures. The wireless control device **120** is assigned an Internet Protocol (IP) address upon connecting to the wireless LAN. The wireless control device **120** may store the service set identifier (SSID) and the SSID password of the wireless LAN. After obtaining the IP address, the wireless control device **120** is able to assign an IP address (e.g., different from the IP address of the wireless control device **120**) to the dimmer switch **110**. Alternatively, the dimmer switch **110** may be operable to obtain the IP address from the wireless router **130** using, for example, procedures defined by the Wi-Fi Protected Setup standard.

[0043] The dimmer switch **110** may be associated with (e.g., assigned to) the wireless control device **120**, such that the wireless control device may transmit commands for controlling the intensity of the lighting load **104** or programming the dimmer switch **110**. Such commands may be transmitted to the dimmer switch **110** via the RF signals **106**. Digital messages transmitted to and from the dimmer switch **110** may include, for example, the MAC address and the IP address of the dimmer switch **110**. The dimmer switch **110** is operable to turn the lighting load **104** on and off. The dimmer switch **110** is also operable to adjust the intensity of the lighting load in response to received digital messages, including the MAC address and the IP address of the dimmer switch, for example. In addition, the wireless router **130** may be operable to receive commands for controlling the lighting load **104** from the Internet, and may wirelessly transmit corresponding digital messages to the dimmer switch **110**.

[0044] According to an example embodiment, the dimmer switch **110** may be assigned an IP address, an SSID, an SSID password, and/or a software AP at manufacture, such that the dimmer switch **110** may act as an AP for other communication devices in a LAN. The wireless control device **120** may recognize the dimmer switch **110** as an AP and may connect to the LAN via the dimmer switch **110**. For example, the dimmer switch **110** may connect to router **130** or may perform the functions of the router **130** itself.

[0045] The dimmer switch **110** may also connect to the wireless LAN to discover other dimmer switches (not shown). The dimmer switch **110** may discover the other dimmer switches using any discovery protocol, such as Bonjour, Simple Service Discovery Protocol (SSDP), Bluetooth® Service Discovery Protocol (SDP), DNS service discovery (DNS-SD), Dynamic Host Configuration Protocol (DHCP), Internet Storage Name Service (iSNS), Jini for Java objects, Service Location Protocol (SLP), Session Announcement Protocol (SAP) for RTP sessions, Simple Service Discovery Protocol (SSDP) for Universal Plug and Play (UPnP), Universal Description Discovery and Integration (UDDI) for web services, Web Proxy Autodiscovery protocol (WPAD), Web Services Dynamic Discovery (WS-Discovery), XMPP Service Discovery (XEP-**0030**), and/or XRDS for XRI, OpenID, OAuth, etc. Upon the dimmer switch **110** discovering one or more other dimmer switches, the dimmer switch may create a peer-to-peer network of dimmer switches capable of communicating with one another. For example, the dimmer switches may communicate programming and/or control instructions received from the wireless control device **120**.

[0046] The wireless control device **120** may control the lighting load **104** by communicating instructions to the dimmer switch **110** via the RF signals **106** that cause the dimmer switch **110** to execute control instructions that have been pre-programmed on the dimmer switch **110**. For example, the dimmer switch **110** may be pre-programmed at manufacture or via an update to execute the control instructions. The control instructions may include pre-configured settings (e.g., protected or locked lighting presets), instructions for raising/lowering lighting level, instructions for fading, instructions for scheduling, instructions for turning lights on/off, or any other pre-programmed instruction, for example.

[0047] The wireless control device **120** may also program the settings (i.e., the operating parameters) of the dimmer switch **110** (e.g., when the dimmer switch is in programming mode). For example, the dimmer switch **110** may be a dimmer switch that may have a limited user interface (UI) or may not have any user interface. As such, the user interface of the wireless control device **120** may be used to program the dimmer switch **110**. For example, various wireless communication links described herein, e.g., Wi-Fi signals, optical signals, near field communication (NFC) signals, or proprietary-protocol RF signals, may be used to program any of a number of programmable features provided by the dimmer switch **110**. Such features may be selected via the wireless control device **120**. For example, the wireless control device **120** may program the dimmer switch **110** with such features as protected or locked presets, high-end trim, low-end trim, adjustable delay, fade time, load type, performing communications via wireless communication modes (e.g., as described herein), or being compatible with different lamps. In addition, the wireless control device **120** may be operable to program the dimmer switch **110** to change between modes of operation, for example, between a switching mode, a dimming mode, and/or an electronic timer mode (i.e., a countdown timer mode). The programming signal may be a one-way or two-way serial communication with the dimmer switch **110**.

[0048] A protected preset is a feature that allows the user to lock the present light intensity level as a protected preset lighting intensity to which the dimmer may set the lighting load **104**. For example, when the dimmer switch **110** is turned on while a protected preset is disabled, the dimmer may set the lighting load **104** to the intensity level at which the dimmer was set when the lighting load was last turned off. When the dimmer switch **110** is turned on while protected preset is enabled, the dimmer may set the lighting load **104** to the protected preset intensity level, for example. The protected preset value may be user-programmed. For example, the user may select a

value from among a plurality of allowable values for the protected preset light intensity level. When the lighting load **104** is turned on with protected preset enabled, a processor or controller may access a memory in the dimmer switch **110** to retrieve the user-selected value, and cause the lighting load **104** to be set to the intensity level represented by that value.

[0049] High-end trim (i.e., high-end intensity) is a feature that governs the maximum intensity level to which the lighting load **104** may be set by the dimmer switch **110**. Values for the high-end trim may range between about 60% and about 100% of full intensity, for example. In an example embodiment, the high-end trim may be pre-programmed to be about 90% of full intensity. In a dimmer switch **110**, high-end trim is a feature that may be user-programmed as described herein.

[0050] Similarly, low-end trim (i.e., low-end intensity) is a feature that governs the minimum intensity level to which the lighting load **104** may be set by the dimmer switch **110**. Values for the low-end trim may range between about 1% and about 20% of full intensity, for example. In an example embodiment, the low-end trim may be preprogrammed to be about 10% of full intensity. In a dimmer switch **110**, low-end trim is a feature that may be user-programmed as described herein.

[0051] Delay-to-off is a feature that causes the lighting load **104** to remain at a certain intensity level for a prescribed period of time before fading to off. Such a feature may be desirable in certain situations, such as, for example, when a user wishes to turn out bedroom lights before retiring, but still have sufficient light to make his way safely to bed from the location of the dimmer switch **110** before the lights are completely extinguished. Similarly, the night staff of a large building may wish to extinguish ambient lights from a location that is some distance away from an exit, and may wish to delay the fade to off for a period of time sufficient for them to walk safely to the exit. Delay-to-off times may range from about 10 seconds to about 60 seconds for example. The delay-to-off time may be user-programmed, as described herein. For example, the user may select a value from among a plurality of allowable values for the delay-to-off time. When the lighting load is turned off with the delay-to-off feature enabled, the dimmer switch **110** may access the user-selected value of delay-to-off feature from memory. The lighting load **104** may remain at the current intensity level for a time represented by the user-selected value of delay-to-off feature.

[0052] Fading is a feature whereby the dimmer causes the lighting load **104** to change from one intensity level to another at a certain rate or plurality of successive rates based on different closures of the toggle switch or indicated in the instructions received from the wireless control device **120** and depending on the state of lighting load **104**. Examples of fading are described in greater detail in the **919** patent. U.S. Pat. No. 7,071,634, issued Jul. 4, 2006, entitled LIGHTING CONTROL DEVICE HAVING IMPROVED LONG FADE OFF, discloses a lighting control device that is capable of activating a long fade off from any light intensity and is incorporated herein by reference. Any or all of the features that define the fade features may be user-programmed via the wireless control device **120**.

[0053] Another feature that may be programmed as described herein is load type. The load type may be inductive, resistive, or capacitive. Forward phase-controlled dimming may be desirable where the load is inductive or resistive; reverse phase-controlled dimming may be desirable where the load is capacitive. Thus, the load type may be defined, at least in part, by a feature having a value associated with either forward phase control or reverse phase control.

[0054] In addition, the dimmer switch **110** may comprise an occupancy sensor or may be responsive to a remote occupancy sensor, and may store operating parameters, such as an occupancy sensor sensitivity setting or timeout value that may be programmed by the wireless control device **120**. The wireless control device **120** may also be operable to program the dimmer switch **110** to operate in one of an occupancy mode and a vacancy mode. In the occupancy mode, the dimmer switch **110** operates to turn a controlled lighting load on and off in response to the occupancy sensor. In the vacancy mode, the dimmer switch **110** operates to only turn the lighting load off in response to the occupancy sensor. Examples of occupancy and vacancy sensors are

described in greater detail in commonly-assigned U.S. Pat. No. 7,940,167, issued May 10, 2011, entitled BATTERY-POWERED OCCUPANCY SENSOR; U.S. Pat. No. 8,009,042, issued Aug. 30, 2011, entitled RADIO-FREQUENCY LIGHTING CONTROL SYSTEM WITH OCCUPANCY SENSING; and U.S. Patent Application No. 8,199,010, issued Jun. 12, 2012, entitled METHOD AND APPARATUS FOR CONFIGURING A WIRELESS SENSOR, the entire disclosures of which are hereby incorporated by reference.

[0055] FIG. 2 is an example screenshot **200** that may be provided on the wireless control device **120** when executing a product control application. The screenshot **200** includes a name field **210** for displaying a name of the lighting load **104** presently being controlled and an intensity field **212** for displaying the present intensity of the controlled lighting load **104**. The wireless control device **120** displays a plurality of soft buttons and controls for the user to actuate to control the lighting load **104**. The same controls may be implemented using hard buttons that correspond to items in the wireless control device **120** display.

[0056] As shown in FIG. 2, the wireless control device **120** displays an on button **220** for turning the lighting load **104** on to the maximum intensity (e.g., approximately 100%), an off button **222** for turning the lighting load **104** off, a raise button **224** for raising the intensity of the lighting load **104** by a predetermined amount, and a lower button **226** for lowering the intensity of the lighting load **104** by a predetermined amount. In addition, the wireless control device **120** displays a virtual slider control **230** having an actuator knob **232** positioned along an elongated vertical slot **234**. The user may touch the actuator knob **232** and slide the knob up and down to respectively raise and lower the intensities of the lighting load **104**. The wireless control device **120** additionally displays a scroll bar **236** that is moved horizontally to cause the wireless control device **120** to control other lighting loads **104** that may be a part of the lighting control system **100**.

[0057] In addition to, or alternative to, the soft buttons illustrated in FIG. 2, the wireless control device **120** display may enable user control of the lighting load **104** via text boxes (e.g., direct entry as a percentage of the maximum intensity), drop boxes, checkboxes, radio buttons, or voice activation. In another example embodiment, the user may control the lighting load **104** by tapping or pressing the wireless control device **120** display (e.g., by tapping or holding the wireless device **120** display to increase or decrease the lighting load **104**). The wireless control device **120** display may be used to select the areas (e.g., rooms), lighting units (e.g., lamps), or dimmer switches that the user wishes to control with the wireless control device **120**. According to another example embodiment, the control device **120** display may include options for the wireless communication link or RF signals **106** upon which the user may wish to communicate with the dimmer switch **110**. The user may set preferences for the type of wireless communication link or RF signals **106** upon which the wireless control device **120** communicates with the dimmer switch **102** based on various factors associated with one or more wireless communication links, e.g., cost, response time, error rate, reliability, etc. For example, if one wireless communication link is more reliable than another, the wireless control device **120** may communicate over the more reliable wireless communication link, if available.

[0058] FIG. 3 is an example screenshot **300** that may be provided on the wireless control device **120** upon entering the programming mode for programming the dimmer switch **110**. Upon entering the programming mode, the wireless control device **120** may transmit a signal to the dimmer switch **110** to put the dimmer switch **110** into programming mode. While in programming mode, the wireless control device **120** may provide various options for allowing a user to select features to be programmed on the dimmer switch **110**. For example, the features may be displayed to a user via the feature display **302**. The feature display **302** may include drop-down boxes, text boxes, soft buttons, radio buttons, checkboxes, or the like, that may allow a user to enter or select one or more features that the user wishes to program on the dimmer switch **110**. The one or more features may be displayed as options (e.g., a list of features) for being programmed, or they may be recognized by the wireless control device **120** upon receipt of entry from a user (e.g., via a text box). The user

may select the one or more features for programming by selecting the enter/select feature button **304**. When the user selects the one or more features, the wireless control device **120** may transmit instructions to the dimmer switch **110** that cause the dimmer switch **110** to be programmed for performing the selected one or more features. For example, the instructions themselves may include software that enables the dimmer switch **110** to perform the selected features, or the instructions may trigger the dimmer switch **110** to retrieve the software from an external source, such as an external server for example.

[0059] The user may exit programming mode by selecting the exit button **306**. By exiting the programming mode, the wireless control device **120** may return to other operating modes and/or transmit a signal to the dimmer switch **110** that returns the dimmer switch **110** to its normal operating mode. According to another example embodiment, the wireless control device **120** may exit the programming mode after a prescribed timeout period in which the wireless control device receives no input commands from the user.

[0060] FIG. **4** is a simplified block diagram of the dimmer switch **110**. The dimmer switch **110** comprises a controllably conductive device **410** coupled in series electrical connection between the AC power source **102** and the lighting load **104** for control of the power delivered to the lighting load. The controllably conductive device **410** may comprise a relay or other switching device, or any suitable type of bidirectional semiconductor switch, such as, for example, a triac, a field-effect transistor (FET) in a rectifier bridge, or two FETs in anti-series connection. The controllably conductive device **410** includes a control input coupled to a drive circuit **412**.

[0061] The dimmer switch **110** further comprises a controller **414** coupled to the drive circuit **412** for rendering the controllably conductive device **410** conductive or non-conductive to thus control the power delivered to the lighting load **104**. The controller **414** may comprise a microcontroller, a programmable logic device (PLD), a microprocessor, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any suitable processing device or control circuit. A zero-crossing detector **415** determines the zero-crossings of the input AC waveform from the AC power supply **102**. A zero-crossing may be the time at which the AC supply voltage transitions from positive to negative polarity, or from negative to positive polarity, at the beginning of each half-cycle. The controller **414** receives the zero-crossing information from the zero-crossing detector **415** and provides the control inputs to the drive circuit **412** to render the controllably conductive device **410** conductive and non-conductive at predetermined times relative to the zero-crossing points of the AC waveform.

[0062] The controller **414** receives inputs from mechanical switches **416** that are mounted on a printed circuit board (not shown) of the dimmer switch **110**, and are arranged to be actuated by the toggle actuator **114** and the intensity adjustment actuator **116**. The controller **414** also controls light-emitting diodes **418**, which are also mounted on the printed circuit board. The light emitting diodes **418** may be arranged to illuminate the status indicators **118** on the front surface of the dimmer switch **110**, for example, through a light pipe structure (not shown). The controller **414** is also coupled to a memory **420** for storage of unique identifiers (e.g., the MAC address and the IP address) of the dimmer switch **110**, the SSID and the SSID password of the wireless LAN, instructions for controlling the lighting load **104**, programming instructions for communicating via a wireless communication link, or the like. The memory **420** may be implemented as an external integrated circuit (IC) or as an internal circuit of the controller **414**. A power supply **422** generates a direct-current (DC) voltage V_{subCC} for powering the controller **414**, the memory **420**, and other low-voltage circuitry of the dimmer switch **110**.

[0063] The dimmer switch **110** further includes a wireless communication module **430** for transmitting and receiving the RF signals **106** to and from the wireless control device **120** and/or the wireless router **130**. For example, the wireless communication module **430** may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. When the wireless

communication module **430** comprises a Wi-Fi module, the controller **414** is operable to control the lighting load **104** in response to received digital messages in Wi-Fi packets (i.e., Internet Protocol packets received via the Wi-Fi signals). The wireless communication module **430** may comprise an RF transceiver and an antenna. Examples of antennas for wall-mounted dimmer switches are described in greater detail in U.S. Pat. No. 5,982,103, issued Nov. 9, 1999, and U.S. Pat. No. 7,362,285, issued Apr. 22, 2008, both entitled COMPACT RADIO FREQUENCY TRANSMITTING AND RECEIVING ANTENNA AND CONTROL DEVICE EMPLOYING SAME, the entire disclosures of which are hereby incorporated by reference.

[0064] The dimmer switch **110** further comprises an optical module **440**, such as an optical signal receiving circuit for example. The optical module **440** may be optically coupled to the optical receiver **119**. The optical module **440** may be coupled to the optical receiver **119** on the front surface of the dimmer switch **110**, for example, through a light pipe (not shown), such that the optical module **440** may receive the optical signals from the wireless control device **120** via the light pipe. For example, the optical module **440** may comprise a photodiode (not shown) that is responsive to the optical signals transmitted by the wireless control device **120**. In addition, the photodiode of the optical module **440** may be controlled by the controller **414**, so as to transmit optical signals to the wireless control device **120** (as will be described in greater detail below), for example.

[0065] The wireless device **120** may control the controllably conductive device **410** using the optical signals and/or the digital messages received via the RF signals **106**. According to an example embodiment, the controller **414** may determine the module from which the signals are received, e.g., from the wireless communication module **430** or the optical module **440**, and the controllably conductive device **410** may be controlled based on those signals. The controller **414** may also transmit messages to the wireless control device **120** via optical signals or digital messages transmitted via the RF signals **106**. For example, the controller **414** of the dimmer switch **110** may be used to transmit digital messages to the wireless control device **120** via wireless communication. The digital messages may include alerts and/or feedback and status information regarding the lighting load **104**. The digital messages may also include error messages or indications as to whether the dimmer switch **110** is able to communicate via a wireless communication link or RF signal **106**, for example.

[0066] FIG. 5 is a front view of the wireless control device **120** showing an example location of a portion **240** of the display screen **122** that may be used for transmitting optical signals. The wireless control device **120** may be operable to transmit a digital message via the optical signals by alternating a portion **240** of the display screen **122** between black and white (or two other contrasting colors) to modulate the light output of (i.e., flash) the portion of the display screen. Instead of turning light/dark transitioning of pixels on the display, the backlight may be on/off transitioned for increased bandwidth. While FIG. 5 illustrates a portion **240** of the display screen **122** for transmitting optical signals, other portions of the display screen or the entire display screen may also be used for transmitting the optical signals.

[0067] FIG. 6 is a perspective view showing the display screen **122** of the wireless control device **120** directed towards the optical receiver **119** of the dimmer switch **110** while the wireless control device is transmitting the optical signals. The portion **240** of the display screen **122** may be held close to the optical receiver **119** while the wireless control device **120** is transmitting the optical signals to ensure that the optical module **440** receives the optical signals. The proximity of the wireless control device **120** to the optical receiver **119** may be close enough for successfully transmitting the optical signal based on ambient light, the signal-to-noise ratio, error coding, etc. The wireless control device **120** may detect the dimmer switch **110** for performing optical signal communications. For example, the wireless control device **120** may comprise a proximity detector and may begin transmitting the optical signals to the optical module **440** when the wireless control device detects that the wireless control device is a predetermined distance from the dimmer switch

110.

[0068] According to another example embodiment, the dimmer switch **110** may be equipped with an electrostatic foam stylus tip, which is located a preset distance from the photodiode, for example. The wireless control device **120** may detect the presence of the stylus tip, correlating the touch location with accelerometer data, to yield the specific display location in which to transmit the data for example. An application on the wireless control device **120** may use the built in proximity sensor as well as the stylus tip to ensure the display flashes when touching the dimmer switch **110** to be programmed. Alternatively, the proximity of the wireless control device **120** may be detected by the dimmer switch **110** and an indication may be transmitted to the wireless control device **120** via the digital message described herein. According to another embodiment, the optical module **440** could be located so as to receive the optical signals through the light pipe structure located between the status indicators **118** and the LEDs **418**, such that the separate optical receiver **119** is not required on the front surface of the dimmer switch **110**.

[0069] Additional bandwidth may be achieved via a tricolor red/green/blue (RGB) photodiode receiver assembly for an approximately three-times increase in bandwidth. For example, the portion **240** of the display screen **122** may be changed between different colors (e.g., red, green, and blue) or even more colors. Multiple bits may be encoded into the transitions from one specific color to another. Transfer rates may be as high as 60 bits/sec using the RGB photodiode receiver, for example.

[0070] According to an example embodiment of the present invention, the wireless control device **120** is able to execute the product control application to assign a unique identifier (e.g., the IP address) to the dimmer switch **110**, before associating the wireless control device **120** with the dimmer switch, for example. The wireless control device **120** chooses a unique identifier for the dimmer switch **110** that is different than the unique identifier of the wireless control device **120** or any other devices on the wireless LAN. The unique identifier may be stored at the wireless control device **120**, the dimmer switch **110**, or on any other network entity, for example. The portion **240** of the display screen **122** may be held close to the optical receiver **119** before the wireless control device **120** begins transmitting the unique identifier via optical signals. The display screen **122** may also be used to transmit the SSID and the SSID password to the dimmer switch **110** via the optical signals. When the dimmer switch **110** successfully receives the unique identifier, the SSID, and/or the SSID password from the wireless control device **120**, the dimmer switch **110** may connect to the local wireless network. For example, the dimmer switch **110** may connect to a wireless LAN and transmit the MAC address of the dimmer switch to the wireless control device **120**. The dimmer switch **110** may provide an indication (e.g., blink the lighting load **104** or provide another indication) that the dimmer switch has successfully connected to the wireless LAN. The wireless control device **120** may transmit and receive the RF signals **106** to and from the dimmer switch **110** to control the lighting load **104**.

[0071] According to another alternate embodiment of the present invention, the camera lens **124** and the camera flash light source **126** may be used to transmit and receive optical signals to and from the optical module **440**. FIG. 7 is a perspective view showing the camera lens **124** and the camera flash light source **126** of the wireless control device **120** directed towards the optical receiver **119** of the dimmer switch **110** according to embodiments of the present invention described herein. The camera flash light source **126** (which may comprise an LED) is controlled to generate the optical signals to thus transmit the digital messages to the dimmer switch **110**. In addition, the photodiode of the optical module **440** of the dimmer switch **110** may be controlled to transmit optical signals. The optical signals may be received by the camera lens **124** of the wireless control device **120**. Accordingly, a two-way optical communication link may be established between the dimmer switch **110** and the wireless control device **120**.

[0072] As illustrated in FIG. 8, the display screen **122** may display a targeting image (e.g., cross-hairs **250**). The target image may assist the user in lining up the camera lens **124**, the camera flash

light source **126**, or the portion **240** of the display screen **122** with the optical receiver **119** of the dimmer switch **110**. The display screen **122** may display a message to the user upon making a determination that the camera lens **124** and the camera flash light source **119** are aligned with the optical receiver **119** of the dimmer switch **110**. The determination may be made by the wireless control device **120** itself, or the wireless control device **120** may make the determination based on an indication received from the dimmer switch **110**. The indication may be received via an optical or other wireless signal for example.

[0073] FIG. **9** is a simplified flow diagram illustrating an example embodiment for optically programming or controlling a dimmer switch **110** via a wireless control device **120**. As shown in FIG. **9**, a programming or controlling procedure **900** may be started at **902** and the wireless control device **120** may launch an application at **904**. The application may display a user interface on the visual display **122** of the wireless control device **120**. The application may enable the adjustment of the lighting load **104** or programming of the dimmer switch **110** via the optical signals. For example, the application may determine instructions for controlling or settings for programming the dimmer switch **110** at **806**. Various settings or instructions may be input and/or stored to the wireless control device **120** application which may be transmitted to the dimmer switch **110**, for example, via an optical signal at **908**.

[0074] When the wireless control device **120** application completes the transmission, the dimmer switch **110** and/or the wireless control device **120** may provide an indication that the transmission has been completed. For example, the wireless control device **120** may receive an indication or message from dimmer switch **110** and/or provide an indication (e.g., audio alert, visual alert, or vibration) to a user at **910**. According to another example embodiment, the dimmer switch **110** may display a message at the dimmer display or provide an indication via the lighting load **104** (e.g., blink the lamp associated with the dimmer) when the transmission has been received and/or processed. After the indication has been provided, the programming or controlling procedure **900** may end at **912**.

[0075] According to an example embodiment, the wireless control device **120** and the application may be used to setup custom lighting schedules at the dimmer switch **110**, such as lighting timer schedules, for example. The user interface provided on the display screen **122** of the wireless control device **120** may provide an easy-to-use interface for configuring the timeclock event times and actions of the timeclock schedule. After the timeclock schedule is configured, the wireless control device **120** may transmit (e.g., via the optical signals) the information defining the timeclock schedule to the dimmer switch **110**, which may be stored in the memory **420** in the dimmer switch **110**. In addition, the wireless control device **120** may transmit the present time of day (and possible the present time of year) to the dimmer switch **110** when transmitting the timeclock schedule information to thus calibrate the internal clock of the dimmer switch **110**.

[0076] To transmit the timeclock schedule information, the application may enter a timer program mode and may be placed in close proximity to the dimmer switch **110**. The wireless control device **120** may transmit the schedule data to the dimmer switch **110**, for example, optically via off/on transitions of the display, which may be similar to the low/hi transitions of a standard serial data stream. When data transfer is complete (e.g., checksums match) the dimmer switch **110** may provide an indication (e.g., audio signal beeps). If data transfer is not successful the programming process may repeat (e.g., 1-n times). If the process fails on the nth attempt an error message (e.g., tone) may indicate a failed programming attempt. The dimmer switch **110** may run the schedules as programmed on the control device **120** application. Additionally, the control device **120** may run the schedules informing the user of the next scheduled event.

[0077] According to an example embodiment, the wireless control device **120** and the application may be used to program the dimmer switch **110** for wireless communication (e.g., Wi-Fi communication) via the local area network. The user may use the application to select a desired router or local area network for performing Wi-Fi communications via the dimmer switch **110**. For

example, the user may enter the name of the dimmer switch **110** for communicating on the local area network. The wireless control device **120** application prompts the user to select a program button, at which time the user may place the wireless control device **120** close to or against the dimmer switch **110** for programming via the optical signal. The optical signal may be received at the dimmer switch **110** via the optical module **440**. The wireless control device **120** may transmit the acquired data to the dimmer switch **110**, for example, optically via the black/white transitions of the display. Once the dimmer switch **110** successfully receives the data, the dimmer switch may join the network and obtain an IP address, which may become the static IP address of the dimmer switch. Once the dimmer switch **110** has an IP address, the dimmer switch sends a TCP/IP sockets message that includes the IP address, name, and/or serial number of the dimmer switch to the IP address of the wireless control device **120**. In addition, the dimmer switch **110** may receive the IP address from the wireless control device **120** via the black/white transitions of the display.

[0078] FIG. **10** is a simplified flow diagram illustrating an example embodiment for programming or controlling a dimmer switch **110** using a Wi-Fi signal. As shown in FIG. **10**, a programming or control procedure **1000** may be started at **1002** and the wireless control device **120** may launch an application that displays a user interface and enables Wi-Fi communication of user inputs from the wireless control device **120** to the dimmer switch **110** at **1004**. The wireless communication module **430** of the dimmer switch **110** may comprise a Wi-Fi module, such as a Wi-Fi receiver for example, to enable Wi-Fi communications. At **1006**, the wireless control device **120** may assign a unique identifier (e.g., IP address and/or an SSID and the corresponding password) to the dimmer switch **110**. The wireless control device **120** may transmit the unique identifier to the dimmer switch **110** at **1008**, for example, via optical signals as described above. The application may determine instructions or settings for programming or controlling the dimmer switch **110** at **1010**. Various settings or instructions may be input and/or stored to the wireless control device **120** application which may be transmitted (e.g., via a local area network or point-to-point communication) to the dimmer switch **110** via a Wi-Fi signal at **1012**. When the wireless control device **120** completes the transmission, the dimmer switch **110** and/or the wireless control device **120** may provide an indication that the transmission has been completed. For example, the wireless control device **120** may receive an indication or message from dimmer switch **110** and/or provide an indication (e.g., audio alert, visual alert, or vibration) to a user at **1014**. After the indication has been provided, the programming or controlling procedure **1000** may end at **1016**.

[0079] According to an example embodiment, the Wi-Fi signal may include a non-standard Wi-Fi signal used to communicate via a vendor-specific proprietary access point. In this embodiment, the dimmer switch **110** may receive Wi-Fi communications via a vendor-specific beacon implementing a vendor-specific protocol. Using the vendor-specific beacon, vendor proprietary information may be included in the Wi-Fi signal, for example, as embedded information in a portion of the beacon management frame. The commands may be embedded in the beacon management frame using active and directed probe request/response for example.

[0080] FIG. **11** is a simplified block diagram of a dimmer switch **1110** according to an alternate embodiment of the present invention. The dimmer switch **1110** is identical to the dimmer switch **110** as shown in FIG. **4**. However, the dimmer switch **1110** comprises an NFC module **1140** for receiving NFC signals from the wireless control device **120**. The NFC module **1140** is coupled to a memory **1120** for storing operating parameters of the dimmer switch **1110**. The memory **1120** may comprise, for example, an electrically-erasable programmable memory (EEPROM) that may be written to without the use of power (e.g., part number M24LR64-R manufactured by ST Microelectronics). The NFC module **1140** may also be coupled to a controller **1114**, which may be operable to control the controllably conductive device **410** to thus control the lighting load **104** in response to the NFC signals.

[0081] FIG. **12** is a simplified flow diagram illustrating an example embodiment for programming or controlling the dimmer switch **1110** using NFC signals. As shown in FIG. **12**, a programming or

control procedure **1200** may be started at **1202** and the wireless control device **120** may launch an application at **1204**. The application may display a user interface on the visual display **122** of the wireless control device and enables receipt of NFC signals from the wireless control device **120** to the dimmer switch **1110**. The application may determine instructions or settings for programming or controlling the dimmer switch **1110** at **1206**. Various settings or instructions may be input and/or stored to the wireless control device **120** application which may be transmitted to the dimmer switch **1110** via the NFC signals at **1108**. The wireless control device **120** may be moved close to the dimmer switch **110** for transmission of the NFC signals. When the wireless control device **120** completes the transmission of the NFC signals, the dimmer switch **1110** and/or the wireless control device **120** may provide an indication that the transmission has been completed. For example, the wireless control device **120** may receive an indication or message from dimmer switch **1210** and/or provide an indication to a user at **1210**. After the indication has been provided, the programming or controlling procedure **1200** may end at **1212**.

[0082] According to an alternate embodiment, the NFC signals may be transmitted to the dimmer switch **1110** to program the dimmer switch with operating parameters when an airgap switch (not shown) of the dimmer switch **1110** is opened as shown in the simplified flow diagram of FIG. **13**. As shown in FIG. **13**, a programming procedure **1300** may be started at **1302** and the wireless control device **120** may launch an application at **1304**. The user determines settings for programming the dimmer switch **1110** using the application at **1306** and then opens the airgap switch of the dimmer switch at **1308**, such that the controller **1114** is unpowered. After the airgap switch is opened at **1308**, the wireless control device **120** is moved close to the dimmer switch **1110** for transmission of the NFC signals at **1310**. When the wireless control device **120** completes the transmission of the NFC signals, the wireless control device **120** provides an indication that the transmission was completed at **1312**. The user closes the airgap switch of the dimmer switch **1110** at **1314**, after which the dimmer switch reboots with the new operating parameters at **1316**. The dimmer switch **1110** blinks the lighting load **104** at **1318** if the dimmer switch was updated correctly, before the programming procedure **1300** ends at **1320**.

[0083] Opening the airgap switch during the programming procedure **1300** helps to isolate the dimmer switch **1110** that is being programmed from other dimmer switches that may be installed near that dimmer switch (e.g., ganged with the dimmer switch in the same electrical wallbox), such that the other dimmer switches are not programmed by mistake. For example, the controller **1114** may prevent the NFC module **1140** from writing to the memory **1120** when the controller is powered. However, when the controller **1114** is unpowered, the controller will stop preventing the NFC module from writing to the memory **1120**.

[0084] The dimmer switch **1110** may be programmed, via the wireless control device **120** for example, to the corresponding low-end and high-end intensities that provide for optimum operation of a particular lamp from a particular manufacturer. Since the operation of a lamp can vary from one lamp to the next (particularly for screw-in compact fluorescent lamps and screw-in light-emitting diode lamps), the wireless control device **120** may retrieve the appropriate the low-end and high-end intensities that correspond to a particular lamp. The information may be retrieved by scanning a barcode on a packaging of the lamp (e.g., using a camera of a smart phone) and then reading the low-end and high-end intensities from the memory **1120** or obtaining low-end and high-end intensities via the Internet, for example. After the wireless control device **120** determines the low-end and high-end intensities of the particular lamp from the retrieved information, the control device **120** programs the dimmer with the appropriate low-end and high-end intensities for the particular lamp. Alternatively, the wireless control device **120** may be operable to program only one of the high-end and low-end intensities or another operating parameter after scanning the barcode of the lamp.

[0085] FIG. **14** is a simplified flow diagram of a programming procedure **1400** for programming the low-end and high-end intensities of the dimmer switch **1110** by scanning a barcode of a

particular lamp. The programming procedure **1400** is started at **1402** and the wireless control device **120** launches an application at **1404**. The user then scans the barcode on the packaging of the lamp at **1406** using, for example, the camera of the wireless control device **120**. The application on the wireless control devices **120** obtains the new high-end and/or low-end intensity, for example, from memory or via the Internet at **1408**, and then transmits the new high-end and/or low-end intensity to the dimmer switch **1110** via the NFC signals at **1410**. When the wireless control device **120** completes the transmission of the NFC signals, the dimmer switch **1110** and/or the wireless control device **120** provides an indication that the transmission has been completed at **1412** and the programming procedure **1400** ends at **1114**. While the programming procedure **1400** of FIG. **14** is shown with the wireless control device **120** transmitting NFC signals to the dimmer switch **1110**, the wireless control device **120** could alternatively transmit the new high-end and/or low-end intensity to the dimmer switch using Internet Protocol packets or optical signals as shown and described above.

[0086] According to an example embodiment, the wireless control device **120** application may use the camera lens **124** (FIG. **6**) and a camera flash lighting source **126** (FIG. **6**) to take a photograph of a lamp for programming the dimmer switch **1110**. The dimmer switch **1110** may be programmed with the corresponding low-end and high-end limits for the photographed lamp. For example, the wireless control device **120** may transmit commands to the dimmer switch **1110** to program the lamp with the low-end and high-end intensities. The control device **120** may analyze the photograph and perform a look-up to determine the limits for the photographed lamp. The look-up may be performed locally or by retrieving information from an external source (e.g., external server) using information obtained during the analysis. According to another embodiment, the control device **120** may send the photograph of the lamp to the external source (e.g., external server) to retrieve the low-end and high-end limit information for the photographed lamp.

[0087] FIG. **15** is a simple diagram of an RF lighting control system **1500** comprising a dimmer switch **1510**, a wireless control device **1520**, and a gateway device **1540** according to an alternate embodiment. The wireless control device **1520** may be operable to transmit RF signals **106** including Internet Protocol packets to the gateway device **1540** via the wireless router **130**. The gateway device **1540** is then operable to transmit digital message according to a proprietary RF communication protocol (such as, for example, the Clear Connect™ protocol) to the dimmer switch **1510** via RF signals **1506**. The dimmer switch **1510** includes a wireless communication module operable to receive digital messages according to the proprietary RF communication protocol via the RF signals **1506**. In addition, a communication dongle (not shown) could be connected to the wireless control device **1520** to allow for direct communication between the wireless control device **1520** and the dimmer switch **1510** using the proprietary RF communication protocol. For example, the communication dongle could be plugged into a headphone jack on the wireless control device **1520**.

[0088] FIG. **16** is a simplified flow diagram illustrating an example embodiment for programming or controlling the dimmer switch **1510** using the proprietary RF communication protocol. As shown in FIG. **16**, a programming or control procedure **1600** may be started at **1602** and the wireless control device **1520** may receive the communication dongle at **1604**. At **1606**, the wireless control device **1520** may launch an application for sending user input and instructions using the proprietary RF communication protocol. The dimmer switch **1510** may be placed into a programming mode for receiving the RF signals **1506** according to the proprietary RF communication protocol and the user may select a dimmer from a list on the wireless control device **1520**. The application may determine instructions or settings for programming or controlling the dimmer switch **1510** at **1608**.

[0089] Various settings or instructions may be input and/or stored to the wireless control device **1520** application which may be transmitted to the dimmer switch **1510** settings or instructions may be transmitted using the proprietary RF communication protocol at **1610**. According to an example

embodiment, the transmission at **1608** may be performed via a direct communication between the wireless control device **1520** and the dimmer switch **1510** using the communication dongle. When the wireless control device **1520** application completes the transmission, the dimmer switch **1510** and/or the wireless control device **1520** may provide an indication that the transmission has been completed. For example, the wireless control device **1520** may receive an indication or message from dimmer switch **1510** and/or provide an indication to a user at **1612**. After the indication has been provided, the programming or control procedure may end at **1614**.

[0090] According to an alternative embodiment, the transmission at **1610** may be performed via multiple communications, such as a non-proprietary-protocol communication (e.g., Wi-Fi) between the wireless control device **1520** and the gateway device **1540** and a proprietary-protocol communication from the gateway device **1540** to the dimmer switch **1510**, for example. In this embodiment, the communication dongle may not be used or even received at **1604**.

[0091] FIG. **17** is a simplified flow diagram illustrating an example embodiment for programming the dimmer switch **1510** using the wireless control device **1520** to have an appropriate low-end intensity for a particular lamp. As shown in FIG. **17**, a programming procedure **1700** may be started at **1702** and the wireless control device **120** may launch an application at **1704** that displays a user interface for configuring the dimmer switch **1510** to the corresponding limits for a particular lamp. The dimmer switch **1510** may be placed into a programming mode (e.g., pressing or holding a button on the dimmer switch). The dimmer switch **1510** may repetitively transmit out a “programming mode” digital message that may be received by the wireless control device **1520** at **1706**. The wireless control device **1520** may display the information regarding the dimmer switch **1510** that is received in the programming message on the visual display **122** (e.g., a list of dimmer switches). The wireless control device **1520** receives a user selection of the dimmer switch **1510** from the list of dimmer switches at **1708**.

[0092] The user then points the camera lens **124** (FIG. **6**) of the wireless control device **1520** at the lamp being controlled by the dimmer switch **1510** and actuates a start button displayed on the visual display **122** at **1710**. The wireless control device **1520** may then transmit commands to the dimmer switch **1510** at **1712** to control the intensity of the lamp to a first intensity (e.g., approximately 50%-100%). The dimmer switch **1510** then slowly begins to decrease the intensity of the lamp towards the low-end intensity at **1714**. When the dimmer switch **1510** tries to control the intensity of the lamp below the lowest intensity to which the lamp may be controlled, the lamp may begin to flicker. The application on the wireless control device **1520** may use the camera lens **124** (FIG. **6**) and the camera flash lighting source **126** (FIG. **6**) to detect the flicker in the lamp. When the flicker has been detected at **1716**, the wireless control device **1520** may transmit a command to the dimmer switch **1510** at **1718** to begin slowly increasing the intensity of the lamp. When the wireless control device **1520** detects that the lamp has stopped flickering at **1720**, the wireless control device **1520** may transmit a command to the dimmer switch **1510** at **1722** to stop adjusting the intensity of the lamp. The dimmer switch **1520** then stores the present intensity of the lamp as a new low-end intensity at **1724**, before the programming procedure **1700** ends at **1726**. When the wireless control device **1520** application completes the programming, the dimmer switch **1510** and/or the wireless control device **1520** may provide an indication that the load limit programming has been completed.

[0093] While the present application has been described with reference to the dimmer switches **110**, **1110**, **1510**, and the wireless control devices **120**, **1520**, the concepts of the present invention could be applied to any control devices that are operable to communicate with each other, such as, for example, dimming ballasts for driving gas-discharge lamps; light-emitting diode (LED) drivers for driving LED light sources; screw-in luminaires including integral dimmer circuits and incandescent or halogen lamps; screw-in luminaires including integral ballast circuits and compact fluorescent lamps; screw-in luminaires including integral LED drivers and LED light sources; electronic switches, controllable circuit breakers, or other switching devices for turning appliances

on and off; plug-in load control devices, controllable electrical receptacles, or controllable power strips for each controlling one or more plug-in loads; motor control units for controlling motor loads, such as ceiling fans or exhaust fans; drive units for controlling motorized window treatments or projection screens; motorized interior or exterior shutters; thermostats for a heating and/or cooling systems; temperature control devices for controlling setpoint temperatures of HVAC systems; air conditioners; compressors; electric baseboard heater controllers; controllable dampers; humidity control units; dehumidifiers; water heaters; pool pumps; televisions; computer monitors; audio systems or amplifiers; generators; electric chargers, such as electric vehicle chargers; an alternative energy controllers; occupancy sensors, vacancy sensors, daylight sensors, temperature sensors, humidity sensors, security sensors, proximity sensors, keypads, battery-powered remote controls, key fobs, cell phones, smart phones, tablets, personal digital assistants, personal computers, timeclocks, audio-visual controls, safety devices, and central control transmitters. [0094] Although the present invention has been described in relation to particular embodiments thereof, many other variations and modifications and other uses will become apparent to those skilled in the art. It is preferred, therefore, that the present invention be limited not by the specific disclosure herein, but only by the appended claims. Additionally, the embodiments described herein may be implemented as a set of computer-executable instructions stored on a computer-readable medium, such as a random-access or read-only memory for example. Such computer-executable instructions may be executed by a processor or microcontroller, such as a microprocessor, within the dimmer switch **110** or the wireless control device **120**, for example.

Claims

1. An electric load control apparatus comprising: near field communication (NFC) interface circuitry to communicate using an NFC communication protocol; wireless local area network (WLAN) interface circuitry to communicate using a WLAN communication protocol different from the NFC communication protocol; memory circuitry; and electric load control circuitry to: establish an NFC connection with a remote device via the NFC interface circuitry; receive access credentials for a wireless local area network (WLAN) from the remote device via the NFC connection; establish a WLAN connection via the WLAN interface circuitry using the received wireless credentials; receive at least one electric load control apparatus operating parameter from the remote device via the NFC connection; and cause a storage of the received operating parameter in the memory circuitry.
2. The electric load control apparatus of claim 1, wherein the electric load control circuitry to further: cause the electric load control apparatus to produce at least one user perceptible output responsive to successful establishment of the WLAN connection.
3. The electric load control apparatus of claim 1, wherein the electric load control circuitry to further: receive, via the WLAN connection, a command generated by an application executed by a remote device.
4. The electric load control apparatus of claim 3, wherein to receive the command generated by the application executed by the remote device, the control circuitry to further: receive, via the WLAN connection, a command generated by an application executed by a smartphone.
5. The electric load control apparatus of claim 1 wherein to receive the at least one electric load control apparatus operating parameter, the control circuitry to further: receive, via the NFC connection, at least one electric dimmer switch operating parameter.
6. The electric load control apparatus of claim 5 wherein to receive the at least one electric dimmer switch operating parameter, the control circuitry to further: receive, via the NFC connection, a first input to set the electric dimmer switch to a minimum level of illumination; and receive, via the NFC connection, a second input corresponding to an electric dimmer switch low-end trim operating parameter.

7. A method of configuring an electric load control apparatus, comprising: establishing by electric load control circuitry, an NFC connection with a remote device via the NFC interface circuitry; receiving by the electric load control circuitry, one or more access credentials for a wireless local area network (WLAN) from the remote device via the NFC connection; establishing by the electric load control circuitry, a WLAN connection via the WLAN interface circuitry using the received wireless credentials; receiving by the electric load control circuitry, at least one electric load control apparatus operating parameter from the remote device via the NFC connection; and causing by the electric load control circuitry, a storage of the received operating parameter in the memory circuitry.

8. The method of claim 7, further comprising: causing by the electric load control circuitry, the electric load control apparatus to produce at least one user perceptible output responsive to successful establishment of the WLAN connection.

9. The method of claim 7, further comprising: receiving by the electric load control circuitry via the WLAN connection, a command generated by an application executed by the remote device.

10. The method of claim 9, wherein receiving the command generated by the application executed by the remote device further comprises: receiving by the electric load control circuitry via the WLAN connection, the command generated by an application executed by a smartphone.

11. The method of claim 7 wherein receiving the at least one electric load control apparatus operating parameter further comprises: receiving by the electric load control circuitry via the NFC connection, at least one electric dimmer switch operating parameter.

12. The method of claim 11 wherein receiving the at least one electric dimmer switch operating parameter further comprises: receiving by the electric load control circuitry via the NFC connection, a first input to set the electric dimmer switch to a minimum level of illumination; and receiving by the electric load control circuitry via the NFC connection, a second input corresponding to an electric dimmer switch low-end trim operating parameter.

13. A non-transitory, machine-readable, storage device that includes instructions that, when executed by electric load control circuitry disposed in an electric load control device, cause the control circuitry to: establish an NFC connection with a remote device via the NFC interface circuitry; receive one or more access credentials for a wireless local area network (WLAN) from the remote device via the NFC connection; establish a WLAN connection via the WLAN interface circuitry using the received wireless credentials; receive at least one electric load control apparatus operating parameter from the remote device via the NFC connection; and cause a storage of the received operating parameter in the memory circuitry.

14. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load control circuitry, further cause the electric load control circuitry to: cause the electric load control apparatus to produce at least one user perceptible output responsive to successful establishment of the WLAN connection.

15. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load control circuitry, further cause the electric load control circuitry to: receive, via the WLAN connection, a command generated by an application executed by the remote device.

16. The non-transitory, machine-readable, storage device of claim 15 wherein the instructions that cause the electric load control circuitry to receive the command generated by the application executed by the remote device further cause the electric load control circuitry to: receive, via the WLAN connection, the command generated by an application executed by a smartphone.

17. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions that cause the electric load control circuitry to receive the at least one electric load control apparatus operating parameter further cause the electric load control circuitry to: receive, via the NFC connection, at least one electric dimmer switch operating parameter.

18. The non-transitory, machine-readable, storage device of claim 17 wherein the instructions that

cause the electric load control circuitry to receive the at least one electric dimmer switch operating parameter further cause the electric load control circuitry to: receive, via the NFC connection, a first input to set the electric dimmer switch to a minimum level of illumination; and receive, via the NFC connection, a second input corresponding to an electric dimmer switch low-end trim operating parameter.
